

附录4 重要公式

1 三角函数常用公式

(1) 诱导公式.

角 θ 函数	$\frac{\pi}{2}-\alpha$	$\frac{\pi}{2}+\alpha$	$\pi-\alpha$	$\pi+\alpha$	$\frac{3\pi}{2}-\alpha$	$\frac{3\pi}{2}+\alpha$	$2\pi-\alpha$
$\sin \theta$	$\cos \alpha$	$\cos \alpha$	$\sin \alpha$	$-\sin \alpha$	$-\cos \alpha$	$-\cos \alpha$	$-\sin \alpha$
$\cos \theta$	$\sin \alpha$	$-\sin \alpha$	$-\cos \alpha$	$-\cos \alpha$	$-\sin \alpha$	$\sin \alpha$	$\cos \alpha$
$\tan \theta$	$\cot \alpha$	$-\cot \alpha$	$-\tan \alpha$	$\tan \alpha$	$\cot \alpha$	$-\cot \alpha$	$-\tan \alpha$
$\cot \theta$	$\tan \alpha$	$-\tan \alpha$	$-\cot \alpha$	$\cot \alpha$	$\tan \alpha$	$-\tan \alpha$	$-\cot \alpha$

★小结:

$$\begin{cases} \sin\left(\frac{\pi}{2}\pm\alpha\right)=\cos\alpha, \\ \sin(\pi\pm\alpha)=\mp\sin\alpha, \\ \cos\left(\frac{\pi}{2}\pm\alpha\right)=\mp\sin\alpha, \\ \cos(\pi\pm\alpha)=-\cos\alpha, \end{cases} \text{这8个公式要熟稔于心}$$

注 (1) 如上表所示, 奇变偶不变, 符号看象限 (因任一角度均可表示为 $\frac{k\pi}{2}+\alpha$, $k \in \mathbf{Z}$, $|\alpha| \leq \frac{\pi}{4}$,

故 k 为奇数时得角 α 的异名函数值, k 为偶数时得角 α 的同名函数值, 然后在前面加上一个把角 α 看成锐角时原来函数值的符号).

(2) 三角函数在四个象限中的符号如下表所示.

角 θ 所在象限 函数	第一象限	第二象限	第三象限	第四象限
$\sin \theta$	+	+	-	-
$\cos \theta$	+	-	-	+
$\tan \theta$	+	-	+	-
$\cot \theta$	+	-	+	-

(3) $\sec \alpha$ 和 $\csc \alpha$ 的函数值可由 $\frac{1}{\cos \alpha}$ 和 $\frac{1}{\sin \alpha}$ 得出.

(2) 倍角公式.

$$\sin 2\alpha = 2\sin \alpha \cos \alpha, \cos 2\alpha = \cos^2 \alpha - \sin^2 \alpha = 1 - 2\sin^2 \alpha = 2\cos^2 \alpha - 1,$$

$$\sin 3\alpha = -4\sin^3 \alpha + 3\sin \alpha, \cos 3\alpha = 4\cos^3 \alpha - 3\cos \alpha,$$

$$\tan 2\alpha = \frac{2 \tan \alpha}{1 - \tan^2 \alpha}, \cot 2\alpha = \frac{\cot^2 \alpha - 1}{2 \cot \alpha}.$$

(3) 半角公式.

$$\sin^2 \frac{\alpha}{2} = \frac{1}{2}(1 - \cos \alpha), \cos^2 \frac{\alpha}{2} = \frac{1}{2}(1 + \cos \alpha), (\text{降幂公式})$$

$$\sin \frac{\alpha}{2} = \pm \sqrt{\frac{1 - \cos \alpha}{2}}, \cos \frac{\alpha}{2} = \pm \sqrt{\frac{1 + \cos \alpha}{2}},$$

$$\tan \frac{\alpha}{2} = \frac{1 - \cos \alpha}{\sin \alpha} = \frac{\sin \alpha}{1 + \cos \alpha} = \pm \sqrt{\frac{1 - \cos \alpha}{1 + \cos \alpha}},$$

$$\cot \frac{\alpha}{2} = \frac{\sin \alpha}{1 - \cos \alpha} = \frac{1 + \cos \alpha}{\sin \alpha} = \pm \sqrt{\frac{1 + \cos \alpha}{1 - \cos \alpha}}.$$

(4) 和差公式.

$$\sin(\alpha \pm \beta) = \sin \alpha \cos \beta \pm \cos \alpha \sin \beta, \cos(\alpha \pm \beta) = \cos \alpha \cos \beta \mp \sin \alpha \sin \beta,$$

$$\tan(\alpha \pm \beta) = \frac{\tan \alpha \pm \tan \beta}{1 \mp \tan \alpha \tan \beta}, \cot(\alpha \pm \beta) = \frac{\cot \alpha \cot \beta \mp 1}{\cot \beta \pm \cot \alpha}.$$

(5) 积化和差与和差化积公式.

① 积化和差公式.

$$\sin \alpha \cos \beta = \frac{1}{2}[\sin(\alpha + \beta) + \sin(\alpha - \beta)], \cos \alpha \sin \beta = \frac{1}{2}[\sin(\alpha + \beta) - \sin(\alpha - \beta)],$$

$$\cos \alpha \cos \beta = \frac{1}{2}[\cos(\alpha + \beta) + \cos(\alpha - \beta)], \sin \alpha \sin \beta = \frac{1}{2}[\cos(\alpha - \beta) - \cos(\alpha + \beta)].$$

② 和差化积公式.

$$\sin \alpha + \sin \beta = 2 \sin \frac{\alpha + \beta}{2} \cos \frac{\alpha - \beta}{2}, \sin \alpha - \sin \beta = 2 \sin \frac{\alpha - \beta}{2} \cos \frac{\alpha + \beta}{2},$$

$$\cos \alpha + \cos \beta = 2 \cos \frac{\alpha + \beta}{2} \cos \frac{\alpha - \beta}{2}, \cos \alpha - \cos \beta = -2 \sin \frac{\alpha + \beta}{2} \sin \frac{\alpha - \beta}{2}.$$

(6) 万能公式.

$$\text{若 } u = \tan \frac{x}{2} (-\pi < x < \pi), \text{ 则 } \sin x = \frac{2u}{1+u^2}, \cos x = \frac{1-u^2}{1+u^2}.$$

2 一元二次方程基础

① 一元二次方程 $ax^2 + bx + c = 0 (a \neq 0)$.

② 判别式 $\Delta = b^2 - 4ac$.

$$\Delta \geq 0, \text{ 方程有两个实根 } x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}; \Delta < 0, \text{ 方程有两个共轭的复根 } x_{1,2} = \frac{-b \pm \sqrt{4ac - b^2}i}{2a}.$$

③ 根与系数的关系 (韦达定理) $x_1 + x_2 = -\frac{b}{a}, x_1 x_2 = \frac{c}{a}$.

3 因式分解公式

① $(a+b)^2 = a^2 + 2ab + b^2$.

② $(a-b)^2 = a^2 - 2ab + b^2$.

③ $(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$.

④ $(a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$.

⑤ $a^2 - b^2 = (a+b)(a-b)$.

⑥ $a^3 - b^3 = (a-b)(a^2 + ab + b^2)$.

⑦ $a^3 + b^3 = (a+b)(a^2 - ab + b^2)$.

⑧ $a^n - b^n = (a-b)(a^{n-1} + a^{n-2}b + \cdots + ab^{n-2} + b^{n-1})$ (n 是正整数) .

⑨ n 为正奇数时, $a^n + b^n = (a+b)(a^{n-1} - a^{n-2}b + \cdots - ab^{n-2} + b^{n-1})$.

⑩ 二项式定理

$$(a+b)^n = \sum_{k=0}^n C_n^k a^{n-k} b^k = a^n + na^{n-1}b + \frac{n(n-1)}{2!} a^{n-2}b^2 + \cdots + \frac{n(n-1)\cdots(n-k+1)}{k!} a^{n-k}b^k + \cdots + nab^{n-1} + b^n.$$

4 阶乘与双阶乘

① $n! = 1 \cdot 2 \cdot 3 \cdots n$, 规定 $0! = 1$.

② $(2n)!! = 2 \cdot 4 \cdot 6 \cdots (2n) = 2^n \cdot n!$.

③ $(2n-1)!! = 1 \cdot 3 \cdot 5 \cdots (2n-1)$.